



AJAY PRASATH P 2024-CSE ▾

A2**Started on** Wednesday, 20 August 2025, 10:46 AM**State** Finished**Completed on** Wednesday, 20 August 2025, 11:04 AM**Time taken** 18 mins 7 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int c=0;
3  void Factor (int num){
4  {
5      for(int i=1;i<=num;++i){
6          c++;
7          c++;
8          if(num%i==0){
9              c++;
10         }
11     }
12 }
13 }
14 int main(){
15     int num;
16     scanf("%d",&num);
17     Factor(num);
18     printf("%d",c+1);
19     return 0;
20 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)